Section C – Mini Capstone Project

Case Study: Nexus Innovations – Diagnosing Random System Reboots

1. Identify three likely hardware components responsible for the random system reboots.

The three likely hardware components are:

1. **CPU (Processor):** Under heavy workload, the CPU temperature may increase. If it becomes too hot, the system may restart automatically to prevent hardware damage.
2. **Power Supply Unit (PSU):** The PSU may not provide enough stable power to the system when all components are under heavy load. This can cause unexpected restarts.
3. **Cooling System:** Faulty fans, blocked air vents, or poor airflow can cause excessive heat inside the workstation.

2. Why do the workstations operate normally in the morning but restart randomly in the afternoon?

In the morning, the workstations may be relatively cool because they have not been running under heavy workloads for long.

During the afternoon, rendering tasks put more load on the CPU and other components. Their temperatures increase over time. Poor ventilation inside the desk cabinets prevents hot air from escaping, causing overheating.

As a result, the system may restart automatically to protect its hardware.

3. How can stress testing help confirm whether the issue is related to overheating or power supply problems?

Stress testing puts the CPU and other hardware components under heavy load to check their stability.

During the test, technicians can monitor CPU temperature, system stability, and power-related behaviour.

- If the temperature becomes too high and the system restarts, overheating may be the cause.
- If the system restarts under heavy load without excessive temperatures, an insufficient or faulty PSU may be responsible.

Stress testing helps identify how the workstation behaves under demanding conditions.

4. Suggest three remediation steps to prevent random system reboots.

1. **Improve Airflow:** Keep the desk cabinets properly ventilated and ensure that air vents are not blocked.
2. **Maintain the Cooling System:** Clean dust from fans and heatsinks, check that all fans are working, and replace faulty cooling components.
3. **Check Power Supply:** Test the PSU and replace it with a suitable, reliable unit if it cannot provide stable power.

5. Why might upgrading the PSU be necessary in this scenario?

Rendering workstations require more power when CPU and other hardware components operate under heavy workloads.

If the existing PSU cannot provide sufficient and stable power, the system may restart unexpectedly.

Upgrading to a reliable PSU with adequate wattage and the correct power connectors can provide stable power to all components and improve system reliability under heavy workloads.